BUSINESS RECOMMENDATIONS: CUSTOMER RETENTION CAMPAIGNS

Causal inference analysis of churn reduction.

Executive Summary

This report details the findings of a causal inference analysis conducted on the IBM Telco Customer Churn Dataset to evaluate the effectiveness of customer retention campaigns. The objective was to determine if retention offers causally reduce customer churn and identify key customer segments for targeted marketing to maximize Return on Investment (ROI). While the analysis aimed to quantify the impact of these campaigns, the results indicate that the current retention offers, as implemented, have not demonstrated a significant reduction in churn, leading to a net financial loss. This document outlines the findings, discusses the implications, and provides recommendations for optimizing future retention strategies.

Analysis Results and Interpretation

The causal inference analysis employed several methods to estimate the Average Treatment Effect (ATE) of customer retention campaigns on churn.

- **Naive ATE:** A naive estimation yielded an ATE of 0.1564. This unadjusted figure likely overstates the true effect due to confounding factors.
- **Propensity Score Matching (PSM) ATT:** The Average Treatment effect on the Treated (ATT) estimated using PSM was 0.0139. This suggests that for customers who received the retention offer, the reduction in churn was approximately 1.39%, accounting for observed confounders.
- **Inverse Probability Weighting (IPW) ATE:** The IPW method estimated an ATE of 0.0067. This result indicates a smaller, but still potentially positive, impact of the offers on reducing churn across the entire customer base.
- **Doubly Robust ATE:** The Doubly Robust ATE, which combines elements of both PSM and IPW, provided the most conservative estimate at 0.0048. This suggests that the causal effect of the retention offers on reducing churn is very small, estimated at approximately 0.48%.

These results, particularly the Doubly Robust ATE, indicate that the observed reduction in churn attributable to the retention campaigns is minimal. This suggests that the current campaign design or the offers themselves may not be sufficiently compelling to influence customer decisions to stay with the company.

Financial Implications:

The financial outcomes derived from these analyses are stark:

- **Estimated Annual Revenue Saved:** $0.00
- **Campaign Cost:** $25,960.00
- **Net Gain:** -$25,960.00
- **Estimated ROI:** -100.00%

This demonstrates that the current retention campaigns have incurred significant costs without generating any measurable savings in revenue from retained customers. The ROI of -100% highlights a critical need for strategic adjustment.

Most Responsive Customer Segment

Despite the overall ineffectiveness, the analysis identified a specific customer segment that showed a slightly higher propensity to respond to retention efforts:

Customers with tenure greater than 60 months and belonging to the highest monthly charge quartile (Q4).

These long-term, high-value customers are the most likely to be influenced by retention offers. This suggests that while the current offers are not broadly effective, there is a segment of high-value customers who *could* be retained with more targeted and potentially higher-value incentives.

Recommendations

Based on the analysis, the following recommendations are proposed to improve the effectiveness and ROI of customer retention campaigns:

1. **Re-evaluate Offer Strategy:** The current offers may not be attractive enough. Consider:
    - **Personalization:** Tailor offers based on individual customer behavior, value, and churn drivers identified through more granular analysis.
    - **Value-Added Services:** Instead of discounts, explore offering premium features, upgraded plans, or exclusive content for a limited period.
    - **Tiered Incentives:** Develop a tiered system of offers that provides more significant benefits to higher-value, longer-tenured customers.
2. **Targeted Campaign Focus:** Prioritize marketing efforts and resources on the identified high-value segment (tenure > 60 months, monthly charge Q4). Design specific, high-impact retention packages for these customers.
3. **Improve Churn Prediction Models:** Enhance predictive models to better identify customers at *imminent* risk of churning, allowing for proactive and more timely interventions.
4. **A/B Testing and Continuous Monitoring:** Implement rigorous A/B testing for any new retention campaign strategies. Continuously monitor campaign performance, using causal inference methods to measure true effectiveness and iterate on strategies.
5. **Cost-Benefit Analysis Refinement:** Before launching new campaigns, conduct thorough cost-benefit analyses, considering not just the cost of the offer but also the potential lifetime value of

the retained customer. Focus on campaigns with a clear path to positive ROI.

6. **Explore Alternative Retention Drivers:** Investigate other factors that might influence churn beyond price, such as customer service interactions, network quality, and competitor offerings. Retention efforts should address these underlying issues.

Conclusion

The causal inference analysis reveals that the current customer retention campaigns are not cost-effective and have a negligible impact on reducing overall churn. However, the identification of a high-value customer segment suggests an opportunity for more focused and potentially successful interventions. By refining the offer strategy, prioritizing targeted campaigns, and continuously monitoring performance through robust analytical methods, the company can move towards retention strategies that deliver a positive financial return and strengthen customer loyalty.